

CLARESHOLM, AB, Canada

WMO: 712340

Lat: 50.0036N

Lon: 113.6406W

Elev: 1009

StdP: 89.78

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.4	-25.8	-33.1	0.2	-29.2	-29.3	0.3	-25.7	17.6	7.2	14.3	5.2	1.5	0	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	30.7	15.8	28.7	15.3	26.7	14.8	18.3	25.8	17.2	25.0	16.1	24.1	4.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	12.6	21.6	14.3	11.5	20.4	13.1	10.6	18.8	56.0	25.9	52.1	24.8	48.8	24.1	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.5	9.1	-34.9	34.1	3.5	1.9	-37.4	35.5	-39.5	36.6	-41.5	37.6	-44.0	39.0		
(5)			-35.0	20.9	3.5	2.5	-37.5	22.7	-39.6	24.2	-41.5	25.6	-44.1	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-6.4	-5.5	-1.0	4.7	10.1	14.1	17.7	16.8	12.1	5.7	-1.1	-6.1
	DBStd	10.74	9.38	8.57	7.92	5.06	4.38	3.19	3.00	3.27	4.47	5.35	7.79	8.80
	HDD10.0	2525	508	435	344	167	53	4	1	1	28	149	335	500
	HDD18.3	4867	766	668	601	409	256	131	48	67	189	391	583	758
	CDD10.0	756	0	0	1	9	56	128	238	213	93	17	2	0
	CDD18.3	57	0	0	0	0	1	4	27	21	3	0	0	0
	CDH23.3	1420	0	0	0	4	64	113	544	514	176	5	0	0
	CDH26.7	426	0	0	0	0	10	21	177	171	47	0	0	0
Wind	WSAvg	3.5	3.7	3.4	3.8	3.8	3.7	3.5	3.2	3.0	3.2	3.6	3.9	3.6
Precipitation	PrecAvg	420	10	12	21	29	64	95	50	45	44	21	19	15
	PrecMax	753	23	35	64	60	130	368	146	101	135	76	51	32
	PrecMin	179	0	0	1	4	0	13	0	0	5	3	2	0
	PrecStd	121	6	8	14	17	35	72	37	26	29	16	12	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.9	17.7	23.3	28.2	29.5	33.3	33.2	30.7	23.8	18.1	12.6
		MCWB	5.4	5.6	7.5	10.3	13.5	17.1	17.0	16.0	14.3	11.4	8.7	5.7
	2%	DB	10.0	10.8	14.7	19.9	25.1	26.4	30.9	30.7	27.9	20.4	14.3	9.5
		MCWB	4.1	4.2	6.4	9.0	12.4	14.9	16.4	15.3	13.5	10.2	6.9	3.9
	5%	DB	8.0	8.2	12.2	16.9	22.3	24.2	28.5	28.5	25.2	17.3	11.7	7.5
		MCWB	3.1	2.7	5.1	7.6	11.3	14.1	16.3	15.1	12.9	8.9	5.6	2.6
	10%	DB	6.2	6.1	9.8	14.1	19.7	22.3	26.3	26.3	22.2	14.6	9.3	5.6
		MCWB	1.9	1.3	3.8	6.2	10.3	13.0	15.8	14.7	11.9	7.4	3.9	1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	6.2	8.5	11.1	14.7	19.3	20.4	18.7	16.0	12.7	9.7	6.3
		MCDB	11.4	12.2	16.2	21.7	24.5	25.7	28.0	26.6	26.5	21.4	16.2	11.6
	2%	WB	4.6	4.5	6.8	9.5	13.1	16.7	18.9	17.4	14.6	10.7	7.5	4.4
		MCDB	9.3	10.4	13.6	18.5	23.0	23.4	26.1	26.0	24.8	19.0	13.8	8.9
	5%	WB	3.3	2.9	5.5	8.1	12.0	15.2	17.8	16.5	13.4	9.1	5.8	2.9
		MCDB	7.6	8.0	11.9	16.3	20.8	22.2	25.4	25.3	23.3	16.5	11.3	7.2
	10%	WB	2.0	1.5	3.9	6.6	10.8	14.0	16.7	15.5	12.3	7.7	4.2	1.4
		MCDB	6.1	5.8	9.5	13.5	18.7	20.4	24.2	24.0	21.1	14.0	9.1	5.5
Mean Daily Temperature Range	5% DB	MDBR	12.3	12.6	12.5	13.1	13.6	13.0	15.1	15.9	15.2	13.5	11.9	12.1
		MCDBR	13.0	14.7	16.1	18.9	19.1	17.2	19.4	20.9	21.6	17.7	14.4	12.6
		MCWBR	8.6	9.2	9.2	9.8	8.7	8.4	8.4	8.5	9.5	9.4	8.7	8.3
	5% WB	MCDBR	12.7	14.9	15.9	17.5	16.9	15.4	16.6	17.3	19.2	16.8	14.3	12.7
		MCWBR	8.8	9.7	9.5	9.3	7.9	8.4	8.4	7.8	8.9	9.3	8.8	8.7
Clear-Sky Solar Irradiance	taub	0.255	0.262	0.285	0.328	0.337	0.333	0.337	0.352	0.314	0.283	0.260	0.235	
	taud	2.440	2.473	2.466	2.355	2.391	2.445	2.448	2.393	2.478	2.535	2.496	2.453	
	Ebn at Noon	817	904	934	918	920	924	914	882	888	865	806	797	
	Edh at Noon	59	74	89	112	113	108	106	107	88	69	55	50	
All-Sky Solar Radiation	RadAvg	1.21	2.20	3.46	4.79	5.70	6.28	6.92	5.54	3.92	2.43	1.33	0.94	
	RadStd	0.10	0.18	0.28	0.33	0.45	0.44	0.38	0.35	0.35	0.26	0.11	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page